

# Sanghyun Jo

Principal AI Researcher, OGQ · Affiliated Researcher, SNU AIBL

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🎓 Google Scholar    🔗 LinkedIn    📖 OpenReview    🆔 ORCID



## Research Highlights

**Peer-Reviewed Conference Papers**    12 total • 9 as (co-)first author (2 also co-corresponding) • 3 as co-author

- **2026** (6): ICLR (**Oral, Top 1.13%**) (co-first), ACCV (**Oral**) (first author; also NeurIPS 2026 Workshop), ICML (co-author), ECCV ×2 (co-first & co-corresponding), MICCAI (co-author)
- **2025** (2): ICCV (co-first), ICCV (co-author)
- **2024** (2): ECCV (first author), ECCV (co-first)
- **2023** (1): ICCV (first author)
- **2021** (1): ICIP (first author)

**Papers Under Review**    3 total • 1 as first & co-corresponding author • 1 as co-corresponding author • 1 as co-author

**Academic Service**    11 conferences • 2 journals • continuous since 2022

- **Outstanding Reviewer, CVPR 2025** (top 5.6% of 12,593 reviewers)
- Reviewer for NeurIPS, ICLR, ICML, AAAI, CVPR, ICCV, ECCV, and IJCV

## Patents & Industry Experience

- 9 years of industrial AI R&D at **OGQ** (2017 – Present), currently **Principal AI Researcher**
- 13 Korean patents (7 granted, 6 filed), including one filed by the **Seoul National University R&DB Foundation** (2026)

## Biography

My research works toward **Physical AI**: vision systems that perceive the physical world, let people interact with what is perceived, and remain deployable under real-world constraints. I pursue three connected axes, treating them not as separate topics but as the perception, generation, and deployment layers of a single pipeline: **segmentation, extending from images to video and 3D**; **interactive segmentation and editing**; and **multimodal distillation and quantization**. Across all three, my approach is to exploit the implicit knowledge already embedded in large-scale foundation models, using generative priors and lightweight interaction to remove the annotation and optimization burdens that make vision systems expensive to build and maintain.

I have been with **OGQ** since 2017, currently serving as the **Principal AI Researcher** and leading industrial AI research and development to maximize tangible real-world impact. In parallel, I have pursued my research under the guidance of **Prof. Kyungsu Kim** since 2021; following his appointment at Seoul National University, I joined the **SNU AIBL Lab** as an **Affiliated Researcher** in September 2024, where I mentor graduate and undergraduate researchers and lead core projects as a co-corresponding author. Integrating industrial R&D leadership with academic rigor, I aim to develop machine learning systems that are both scientifically grounded and practically deployable.

## Research Interests

Toward **Physical AI**: perception, interaction, and efficient deployment of vision systems

- |                                          |                |
|------------------------------------------|----------------|
| • Core & Biomedical Vision AI            | 2025 – Present |
| • Multimodal Distillation & Quantization | 2025 – Present |
| • Interactive Segmentation & Editing     | 2023 – Present |
| • Image, Video & 3D Perception           | 2021 – Present |

## Appointments

|                                                                                   |                     |
|-----------------------------------------------------------------------------------|---------------------|
| <b>Principal AI Researcher</b><br><i>OGQ</i> (Acquired GYN in 2022), Seoul, Korea | Jul. 2017 – Present |
|-----------------------------------------------------------------------------------|---------------------|

|                                                                                                                            |                     |
|----------------------------------------------------------------------------------------------------------------------------|---------------------|
| <b>Affiliated Researcher</b><br><i>SNU AIBL Lab</i> (Adviser: <i>Prof. Kyungsu Kim</i> ), Seoul National University, Korea | Sep. 2024 – Present |
|----------------------------------------------------------------------------------------------------------------------------|---------------------|

## Education

### B.S., Computer Science

Feb. 2026

National Institute for Lifelong Education, Seoul, Korea

Degree conferred by the Minister of Education via the Bachelor's Degree Examination for Self-Education.

## Publications: Conference Papers

12 peer-reviewed papers: 9 as (co-)first author (2 also co-corresponding), 3 as co-author. Venues: ICLR (1 Oral, Top 1.13%), ACCV (1 Oral), ICML (1), ECCV (4), ICCV (3), MICCAI (1), ICIP (1).

- [C1] **Sanghyun Jo**<sup>\*</sup>, Ziseok Lee<sup>\*</sup>, Wooyeol Lee, Jonghyun Choi, Jaesik Park<sup>†</sup>, Kyungsu Kim<sup>†</sup>.  
"TRACE: Your Diffusion Model is Secretly an Instance Edge Detector"  
**ICLR 2026** [Oral, Top 1.13% (223 papers selected out of 19,735 submissions)].  
(2026 Google h5-index: 362, Google Top Publication Rank 2nd for Artificial Intelligence )  
 PDF  OpenReview  Code  Project [Sanghyun Jo: The co-first author<sup>\*</sup>]
- [C2] **Sanghyun Jo**, Seo Jin Lee, Seohyung Hong, Yoorim Gang, Hyeongsu Kim, Hyungseok Seo<sup>†</sup>, Kyungsu Kim<sup>†</sup>.  
"One Click per Cell Type Suffices: Training-free Group Interaction for Cell Instance Segmentation"  
**ACCV 2026** [Oral]; also accepted at **NeurIPS 2026 Workshop** (ASCI; non-archival).  
(2026 Google h5-index: 52, Google Top Publication Rank 17th for Computer Vision )  
 PDF  Code  Project [Sanghyun Jo: The first author]
- [C3] Ziseok Lee<sup>\*†</sup>, Minyeong Hwang<sup>\*</sup>, Wooyeol Lee, **Sanghyun Jo**, Jihyung Ko, Young Bin Park, Jae-Mun Choi, Eunho Yang<sup>†</sup>, Kyungsu Kim<sup>†</sup>. "On the Collapse of Generative Paths: A Criterion and Correction for Diffusion Steering"  
**ICML 2026** [Acceptance Rate: 26.6% (6,352 accepted out of 23,918 submissions)].  
(2026 Google h5-index: 272, Google Top Publication Rank 3rd for Artificial Intelligence )  
 PDF  OpenReview  Code  Project [Sanghyun Jo: The co-author]
- [C4] **Sanghyun Jo**<sup>\*†</sup>, Wooyeol Lee<sup>\*</sup>, Ziseok Lee<sup>\*</sup>, Jonghyun Choi, Jaesik Park, Kyungsu Kim<sup>†</sup>.  
"ISAC: Training-Free Instance-to-Semantic Attention Control for Multi-Instance Generation"  
**ECCV 2026** [Acceptance Rate: 27.5% (2,883 accepted out of 10,473 submissions)].  
(2026 Google h5-index: 262, Google Top Publication Rank 2nd for Computer Vision )  
 PDF  Code  Project [Sanghyun Jo: The co-first<sup>\*</sup> & co-corresponding<sup>†</sup>]
- [C5] **Sanghyun Jo**<sup>\*†</sup>, Donghwan Lee<sup>\*</sup>, Eunji Jung<sup>\*</sup>, Seong Je Oh, Kyungsu Kim<sup>†</sup>. "EraseLoRA: MLLM-Driven Foreground Exclusion and Background Subtype Aggregation for Dataset-Free Object Removal"  
**ECCV 2026** [Acceptance Rate: 27.5% (2,883 accepted out of 10,473 submissions)].  
(2026 Google h5-index: 262, Google Top Publication Rank 2nd for Computer Vision )  
 PDF  Code  Project [Sanghyun Jo: The co-first<sup>\*</sup> & co-corresponding<sup>†</sup>]
- [C6] Hyeongsu Kim, Sihyun Kim, Minyoung Cho, **Sanghyun Jo**, Minhyeong Lee, Kyungsu Kim<sup>†</sup>.  
"GeneRAG: A Retrieval-Augmented Framework for Spatially Resolved Gene Expression Prediction"  
**MICCAI 2026** [Acceptance Rate: 26.3% (1,211 accepted out of 4,601 submissions)].  
(2026 Google h5-index: 98, Google Top Publication Rank 4th for Medical Imaging )  
 PDF  Code  Project [Sanghyun Jo: The co-author]
- [C7] **Sanghyun Jo**<sup>\*</sup>, Seo Jin Lee<sup>\*</sup>, Seungwoo Lee, Seohyung Hong, Hyungseok Seo<sup>†</sup>, Kyungsu Kim<sup>†</sup>.  
"COIN: Confidence Score-Guided Distillation for Annotation-Free Cell Segmentation"  
**ICCV 2025** [Acceptance Rate: 24.0% (2,698 accepted out of 11,239 submissions)].  
(2026 Google h5-index: 256, Google Top Publication Rank 3rd for Computer Vision )  
 PDF  Code  Project [Sanghyun Jo: The co-first author<sup>\*</sup>]
- [C8] Joowon Kim<sup>\*</sup>, Ziseok Lee<sup>\*</sup>, Donghyeon Cho, **Sanghyun Jo**, Yeonsung Jung, Kyungsu Kim<sup>†</sup>, Eunho Yang<sup>†</sup>.  
"Early Timestep Zero-Shot Candidate Selection for Instruction-Guided Image Editing"  
**ICCV 2025** [Acceptance Rate: 24.0% (2,698 accepted out of 11,239 submissions)].  
(2026 Google h5-index: 256, Google Top Publication Rank 3rd for Computer Vision )  
 PDF  Code  Project [Sanghyun Jo: The co-author]

- [C9] **Sanghyun Jo**, Fei Pan, In-Jae Yu, Kyungsu Kim<sup>†</sup>. "DHR: Dual Features-Driven Hierarchical Rebalancing in Inter- and Intra-Class Regions for Weakly-Supervised Semantic Segmentation"  
**ECCV 2024** [Acceptance Rate: 27.9% (2,395 accepted out of 8,585 submissions)].  
 (2026 Google h5-index: 262, Google Top Publication Rank 2nd for Computer Vision )  
 PDF  Code [Sanghyun Jo: The first author]
- [C10] **Sanghyun Jo**<sup>\*</sup>, Soohyun Ryu<sup>\*</sup>, Sungyub Kim, Eunho Yang, Kyungsu Kim<sup>†</sup>.  
 "TTD: Text-Tag Self-Distillation Enhancing Image-Text Alignment in CLIP to Alleviate Single Tag Bias"  
**ECCV 2024** [Acceptance Rate: 27.9% (2,395 accepted out of 8,585 submissions)].  
 (2026 Google h5-index: 262, Google Top Publication Rank 2nd for Computer Vision )  
 PDF  Code [Sanghyun Jo: The co-first author<sup>\*</sup>]
- [C11] **Sanghyun Jo**, In-Jae Yu, Kyungsu Kim<sup>†</sup>. "MARS: Model-agnostic Biased Object Removal without Additional Supervision for Weakly-Supervised Semantic Segmentation"  
**ICCV 2023** [Acceptance Rate: 26.7% (2,160 accepted out of 8,088 submissions)].  
 (2026 Google h5-index: 256, Google Top Publication Rank 3rd for Computer Vision )  
 PDF  Code [Sanghyun Jo: The first author]
- [C12] **Sanghyun Jo**, In-Jae Yu<sup>†</sup>.  
 "Puzzle-CAM: Improved Localization via Matching Partial and Full Features"  
**ICIP 2021** [Acceptance Rate: 45.9% (791 accepted out of 1,722 submissions)].  
 (2026 Google h5-index: 55, Google Top Publication Rank 16th for Computer Vision )  
 PDF  Code [Sanghyun Jo: The first author]

## Conference Papers Under Review

- [U1] **Sanghyun Jo**<sup>†</sup>, Chae Yeon Lim, Donghwan Lee, Sihyun Kim, Soo Ye Kim (Adobe), Kyungsu Kim<sup>†</sup>. "Depth-to-RGB: Repurposing a Frozen Depth Estimator for Geometry-Guided Compositing", *Under Review*.  
 PDF  Project [Sanghyun Jo: The first & co-corresponding<sup>†</sup>]
- [U2] Jihyung Ko<sup>\*</sup>, Eunji Jung<sup>\*</sup>, Hyeongsu Kim, Ziseok Lee, Jae Won Cho, **Sanghyun Jo**<sup>†</sup>, Kyungsu Kim<sup>†</sup>. "PatchGate: Narrowing the Verbalization Gap with Intrinsic Object Inventories in Frozen Vision-Language Models", *Under Review*.  
 PDF [Sanghyun Jo: The co-corresponding<sup>†</sup>]
- [U3] Chae Yeon Lim<sup>\*†</sup>, Jaehyeon Woo<sup>\*</sup>, **Sanghyun Jo**, Kyungsu Kim<sup>†</sup>. "LiftConsensus: Lifting 2D Change Evidence into 3D via Multi-View Spatial Consensus for Retail Change Detection", *Under Review*. [Sanghyun Jo: The co-author]

## Preprint

- [1] **Sanghyun Jo**, In-Jae Yu, Kyungsu Kim<sup>†</sup>. "RecurSeed and EdgePredictMix: Pseudo-Label Refinement Learning for Weakly Supervised Semantic Segmentation", *arXiv preprint arXiv:2204.06754*.  
 PDF  Code [Sanghyun Jo: The first author]

## Awards & Honors

- **Outstanding Reviewer, CVPR 2025** Jun. 2025  
 Selected as top 5.6% (711 of 12,593 reviewers).
- **1st Place, Legal Judgment Prediction AI Challenge** Jul. 2023  
 1st place out of 1,153 participants. Developed case representation and multi-branch modeling for civil court ruling prediction.
- **5th Place, Synthetic Data Object Detection AI Challenge** Jun. 2023  
 5th place out of 1,747 participants. Built a robust vehicle detector under domain shift.

## Academic Service

Reviewer for **11 international conferences** and **2 journals** since 2022, including NeurIPS, ICLR, ICML, AAAI, CVPR, ICCV, ECCV, and IJCV. Recognized as an **Outstanding Reviewer at CVPR 2025** (top 5.6% of 12,593 reviewers).

- **2027:** ICLR, AAAI
- **2026:** ICLR, NeurIPS, CVPR, ECCV, MICCAI, ACCV
- **2025:** CVPR, ICCV
- **2024:** ICPR, ICIP
- **2023:** ICIP
- **2022:** ICML, ICIP

**Journals:** IJCV (2024–2026), IEEE TCSVT (2022)

## Invited Talks & Presentations

- **Yonsei University**, Seoul, Korea Jun. 2026  
Vision AI Already Knows: Understanding the World Without Labels  
 [Slides \(EN\)](#)
- **ICLR 2026**, Rio de Janeiro, Brazil Apr. 2026  
TRACE Oral Presentation (Oral Session 5B: Video and Scene Generation)  
 [Slides \(EN\)](#)  [Watch](#)
- **MODULABS**, Seoul, Korea Dec. 2024  
DHR & TTD Recap: Sharing Our ECCV 2024 Work with Local Researchers  
 [Watch](#)
- **Yonsei University**, Seoul, Korea Dec. 2023  
Advanced AI Developments  
 [Slides \(EN\)](#)
- **Google Developer Groups (GDG)**, Incheon, Korea Oct. 2023  
Strategies for Reducing Labeling Costs in Vision AI  
 [Slides \(KR\)](#)  [Watch](#)
- **MODULABS**, Seoul, Korea Nov. 2022  
Experience Sharing Session: Vision Research  
 [Slides \(KR\)](#)  [Watch](#)
- **MODULABS**, Seoul, Korea Oct. 2022  
RSEPM Paper Presentation  
 [Slides \(EN\)](#)  [Watch](#)

## Patents

*13 Korean patents (7 granted, 6 filed) on segmentation, image editing, and de-identification, including one filed by the Seoul National University R&DB Foundation.*

- [P1] Kyungsu Kim, Sihyun Kim, Hyeongsu Kim, Minhyeong Lee, Minyoung Cho, **Sanghyun Jo**. "Retrieval-Augmented Framework for Spatially Resolved Gene Expression Prediction", Korean Patent Application No. 10-2026-0138802, Jul. 2026.
- [P2] Seungon Bang, Kyungsu Kim, **Sanghyun Jo**. "Confidence Score-Guided Self-Distillation Method for Annotation-Free Instance Segmentation", Korean Patent Application No. 10-2025-0162415, Oct. 2025.
- [P3] Seungon Bang, Kyungsu Kim, **Sanghyun Jo**. "Instruction-Guided Image Editing Method Using Early-Stopped Background Inconsistency Score Evaluation", Korean Patent Application No. 10-2025-0162189, Oct. 2025.
- [P4] Seungon Bang, Kyungsu Kim, **Sanghyun Jo**. "Instance-Aware Edge Map Generation Without Pixel Annotations Using a Diffusion Model and Segmentation Mask Refinement Method Based Thereon", Korean Patent Application No. 10-2025-0162165, Oct. 2025.
- [P5] Kyungsu Kim, In-Jae Yu, **Sanghyun Jo**, Seungon Bang. "Object De-Identification Method Through Segmentation and Server That Performs the Same Method", Korean Patent No. 10-2025-0018458, Feb. 2025.

- [P6] **Sanghyun Jo**, Kyungsu Kim, Seungon Bang. "Method for Synthesizing Virtual Objects for Virtual Advertising Performed by a Computing Device and Computing Device", Korean Patent No. 10-2024-0198877, Dec. 2024.
- [P7] **Sanghyun Jo**, Kyungsu Kim, Seungon Bang. "Method for Background Composition of Error Object Area Performed on a Computing Device and Computing Device for Performing the Same", Korean Patent No. 10-2024-0198644, Dec. 2024.
- [P8] **Sanghyun Jo**, Kyungsu Kim, Seungon Bang. "Dual Feature-Based Image Segmentation Method Performed by a Computing Device and Computing Device for Performing the Same", Korean Patent No. 10-2024-0171236, Nov. 2024.
- [P9] **Sanghyun Jo**, Kyungsu Kim, Seungon Bang. "Method for Image Tagging Based on Image Matching Performed by a Computing Device and Computing Device for Performing the Same", Korean Patent No. 10-2024-0171187, Nov. 2024.
- [P10] Kyungsu Kim, In-Jae Yu, **Sanghyun Jo**, Seungon Bang. "Debiased Segmentation Method and Server That Performs the Same Method", Korean Patent No. 10-2023-0164587, Nov. 2023.
- [P11] Kyungsu Kim, In-Jae Yu, **Sanghyun Jo**, Seungon Bang. "Obtaining Method of Object Mask Information for Public Safety and Server System Performing the Same", Korean Patent Application No. 10-2022-0176164, Dec. 2022.
- [P12] Kyungsu Kim, In-Jae Yu, **Sanghyun Jo**, Seungon Bang. "Method of Acquiring Object Segmentation Information Through Trained Neural Network and Server System Performing the Same Method", Korean Patent Application No. 10-2022-0141458, Oct. 2022.
- [P13] Seungon Bang, **Sanghyun Jo**. "Method for Safety Management by Object Segmentation Based on Deep-Learning and System", Korean Patent No. 10-2021-0164131, Nov. 2021.